LAMBDA VS NORMAL FUNCTIONS

In [1]:
```
def sum1(x,y):
    return x+y
sum1(10,20)
```

Out[1]: 30

In [4]:
```
sum2=lambda x,y: x+y
sum2(10,20)
```

Out[4]: 30

In [6]:
```
list1=[5,0,9,3,6,8]
def doubler(list1):
    list2=[]
    for x in list1:
        list2.append(2*x)
    return list2
doubler(list1)
```

Out[6]: [10, 0, 18, 6, 12, 16]

In [9]:
```
# write the same program using the map in python
list1=[5,0,9,3,6,8]
def doubler(x):
    return 2*x;
# map will take the list1 elements and applies doubler function on each elemen
# map will makesure that doubler function is applied to each and every element
print(list(map(doubler,list1)))
```

```
[10, 0, 18, 6, 12, 16]
```

In [12]:
```
# we can replace the doubler function with lambda function
list1=[5,0,9,3,6,8]
print(list(map(lambda x: x*2,list1)))
```

```
[10, 0, 18, 6, 12, 16]
```

In [14]:
```
# sum of all elements
def sumlist(x):
    total=0
    for i in x:
        total=total+i
    return total
sumlist(list1)
```

Out[14]: 31

In [20]:
```
# to perform above such operations we need to use the reduce function
from functools import reduce
```

```python
print(reduce(lambda x,y: x+y,list1))
```

31

In [ ]: